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Name: _____

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Department/School: _____

SID: _____

University Physics

Final Examination

Kuang Yaming Honors School, Nanjing University

$$\Gamma(\nu) = \int_0^{+\infty} e^{-x} x^{\nu-1} dx, \quad \Gamma(\nu+1) = \nu\Gamma(\nu), \quad \Gamma(1) = 1, \quad \Gamma(1/2) = \sqrt{\pi}, \quad \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}.$$

Select five out of following six problems.

1.(20pts.) Electric charge is a physical property of matter that is relativistic invariant. It means that the amount of electric charge an object carries does not depend on its velocity of movement. Suppose that the electric charge is uniformly distributed on a body, and the electric charge density is ρ_0 when the body is at rest.

(a) When the body moves, will its charge density increase or decrease to the observer at rest?

(b) If the charge-carrying body moves with a constant speed v along the x -axis, what is its charge density to the rest observer?

Solution: (a) The charge density will increase to the observer at rest.

(b) When the charge-carrying body moves with a constant speed v along the x -axis, its length along the x -axis will contract due to the Lorentz contraction

$$l = l_0 \sqrt{1 - \left(\frac{v}{c}\right)^2}$$

and its width and height along y - and z - axes will be unchanged. Since the total electric charge is invariant, therefore

$$\frac{\rho}{\rho_0} = \frac{l_0}{l} = \frac{1}{\sqrt{1 - \left(\frac{v}{c}\right)^2}} = \gamma.$$

Finally, $\rho = \gamma\rho_0$.

2. (20pts.) Consider an ideal gas of total amount of one mole. Suppose that the gas undergoes a process (polytropic process) such that $p\nu^n = C$ where p is the pressure, ν the molar volume, and n and C are both constant ($n \geq 0$, $n \neq 1$).

(a) Show that in the process

$$\frac{dv}{v} = \frac{1}{1-n} \frac{dT}{T}$$

where T is the temperature of the gas.

(b) Find the work W done by the gas if the temperature of the gas changes from initial T_1 to final T_2 in this process.

Solution: (a) Taking logarithm of both sides of the polytropic equation, we have

$$\ln p + n \ln \nu = \ln C.$$

Making derivative of the both sides of above equation, we obtain

$$\frac{dp}{p} + n \frac{d\nu}{\nu} = 0. \quad (1)$$

Taking logarithm of both side of the equation of state $p\nu = RT$, we have

$$\ln p + \ln \nu = \ln T + \ln R.$$

Making derivative of the both sides of above equation, we obtain

$$\frac{dp}{p} + \frac{dv}{v} = \frac{dT}{T}. \quad (2)$$

(2)-(1):

$$(1-n) \frac{dv}{v} = \frac{dT}{T} \quad \text{or} \quad \frac{dv}{v} = \frac{1}{1-n} \frac{dT}{T}. \quad (3)$$

(b) Rewriting (3) as

$$dv = \frac{v}{1-n} \frac{dT}{T},$$

then we have

$$pdv = \frac{pv}{1-n} \frac{dT}{T} = \frac{RT}{1-n} \frac{dT}{T} = \frac{R}{1-n} dT.$$

Therefore,

$$W = \int pdv = \int_{T_1}^{T_2} \frac{R}{1-n} dT = \frac{R(T_2 - T_1)}{1-n}.$$

3. (20pts.) The experiment shows that a superconducting phase transition occurs in a metal when the temperature is lowered to the critical temperature T_c . That is, when the temperature is above T_c ($T > T_c$), the metal is in the normal metal phase and the isobaric heat capacity of the metal is $C_p = \gamma T$ where γ is a constant, and when the temperature is below T_c ($T < T_c$), the metal enters the superconducting phase in which the electrical resistivity disappears and its isobaric heat capacity behaves exponentially with temperature, i.e., in the form of function $C_p = AT \exp(-\alpha T)$ where A and α are both positive constants. The superconducting phase transition is a second order (continuous) phase transition with no latent heat.

(a) Find ΔS_1 the change in entropy of the system in the superconducting phase when its temperature goes from the absolute zero to T_c .

(b) Find ΔS_2 , the change in entropy in the process of the phase transition exactly at T_c .

(c) Find ΔS_3 , the change in entropy of the system in the normal metal phase when its temperature goes from T_c to $2T_c$.

(d) Find ΔS , the total change in entropy of the system when its temperature goes from the absolute zero to $2T_c$.

Solution: (a)

$$\Delta S_1 = \int_0^{T_c} \frac{C_p dT}{T} = A \int_0^{T_c} e^{-\alpha T} dT = A \left[-\frac{e^{-\alpha T}}{\alpha} \right]_0^{T_c} = \frac{A}{\alpha} (1 - e^{-\alpha T_c}).$$

(b) The superconducting phase transition is a second order transition and the entropy is continuous during the transition. Therefore

$$\Delta S_2 = 0.$$

(c)

$$\Delta S_3 = \int_{T_c}^{2T_c} \frac{C_p dT}{T} = \int_{T_c}^{2T_c} \frac{\gamma T dT}{T} = \gamma T_c.$$

$$(d) \quad \Delta S = \Delta S_1 + \Delta S_2 + \Delta S_3 = \gamma T_c + \frac{A}{\alpha} (1 - e^{-\alpha T_c}).$$

4. (20pts.) There exists such a gas that satisfies the relation $p = u/3$ where p and u are the pressure and the internal energy density (internal energy per unit volume) of the gas, respectively. It is also known that the internal energy density u of the gas is the function of temperature T only, i.e., $u = u(T)$.

(a) Show from $dF = -SdT - pdV$ that $F = -uV/3$, where F , S and V are the Helmholtz free energy, entropy and volume of the system, respectively.

(b) Show that $du/u = 4dT/T$ and then $U = aT^4V$ for the gas, where U is the internal energy of the gas and a is a constant.

(c) Find the Gibbs free energy (Gibbs function) $G = U - TS + pV$ of the gas.

Solution: (a) In the isothermal process at temperature T , we have

$$F(T, V) - F(T, 0) = \int_0^V -pdV = \int_0^V -\frac{u(T)}{3}dV = -\frac{u(T)V}{3}.$$

Since the Helmholtz free energy $F(T, V)$ is an extensive quantity, then $F(T, V)$ is proportional to V . Therefore, we have

$$F(T, 0) = 0.$$

Finally
$$F(T, V) = -\frac{u(T)V}{3}.$$

(b) From $dF = -SdT - pdV$, we have

$$S = -\left(\frac{\partial F}{\partial T}\right)_V = \frac{1}{3} \frac{du}{dT} V$$

Since $F = U - TS$, we have

$$U = F + TS = -\frac{uV}{3} + \frac{T}{3} \frac{du}{dT} V = uV, \quad \text{or} \quad \frac{4uV}{3} = \frac{T}{3} \frac{du}{dT} V \quad \text{or} \quad \frac{du}{u} = 4 \frac{dT}{T}.$$

Integrating both sides of above equation, we obtain

$$\ln u = 4 \ln T + \ln a \quad \text{or} \quad u = aT^4.$$

Finally $U = uV = aT^4V$.

(c) The Gibbs free energy of the gas is

$$G = U - TS + pV = F + pV = -\frac{uV}{3} + \frac{u}{3}V = 0.$$

5. (20 pts.) (a) Show that $TdS = C_v dT + T\left(\frac{\partial p}{\partial T}\right)_V dV$.

(b) Find the change in entropy of a mole of the Van der Waals gas when it undergoes an isothermal process from the initial molar volume v_1 to final molar volume v_2 .

(c) Show that $dU = C_v dT + \left[T\left(\frac{\partial p}{\partial T}\right)_V - p\right]dV$.

(d) By using the equation of state of the ideal gas, show that the internal energy U of an ideal gas is independent of its volume V .

Solution: (a) From the fundamental equation of the thermodynamics

$$dU = TdS - pdV,$$

We have

$$dU = T\left[\left(\frac{\partial S}{\partial T}\right)_V dT + \left(\frac{\partial S}{\partial V}\right)_T dV\right] - pdV = T\left(\frac{\partial S}{\partial T}\right)_V dT + \left[T\left(\frac{\partial S}{\partial V}\right)_T - p\right]dV. \quad (1)$$

Thus

$$C_v = \left(\frac{\partial U}{\partial T}\right)_V = T\left(\frac{\partial S}{\partial T}\right)_V. \quad (2)$$

Therefore

$$TdS = T\left[\left(\frac{\partial S}{\partial T}\right)_V dT + \left(\frac{\partial S}{\partial V}\right)_T dV\right] = C_v dT + T\left(\frac{\partial S}{\partial V}\right)_T dV.$$

Using the Maxwell relation

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V \quad (3)$$

(that can be obtained from $dF = -SdT - pdV$), we have

$$TdS = C_V dT + T\left(\frac{\partial p}{\partial T}\right)_V dV.$$

(b) The equation of state of a mole of Van der Waals gas is

$$p = \frac{RT}{v-b} - \frac{a}{v^2}.$$

Thus $\left(\frac{\partial p}{\partial T}\right)_V = \frac{R}{v-b}$. In the isothermal expansion ($dT = 0$),

$$TdS = T\left(\frac{\partial p}{\partial T}\right)_V dv, \quad ds = \frac{R}{v-b} dv, \quad \Delta S = \int_{v_1}^{v_2} \frac{R}{v-b} dv = R \ln \frac{v_2 - b}{v_1 - b}.$$

(c) From (1), (2) and (3), we have

$$dU = C_V dT + \left[T\left(\frac{\partial p}{\partial T}\right)_V - p \right] dV. \quad (4)$$

(d) From (4), we have

$$\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial p}{\partial T}\right)_V - p. \quad (5)$$

Substituting the equation of state of the ideal gas $p = \frac{nRT}{V}$ into above equation, we have

$$\left(\frac{\partial U}{\partial V}\right)_T = T \frac{nR}{V} - p = 0.$$

Therefore, the internal energy U of an ideal gas is independent of its volume V .

6. (20pts.) A system is constituted of N number of nearly independent one-dimensional harmonic oscillators. According to Boltzmann statistics, the distribution of these oscillators in the phase space can be written as

$$\frac{dn}{N} = C \exp\left(-\frac{E(x, p)}{k_B T}\right) dx dp$$

where C is the normalizing coefficient, T is the temperature of the system, k_B is the Boltzmann constant, x and p denote the coordinate and momentum of a oscillator, and $E(x, p)$ is the mechanical energy of a oscillator that is given by

$$E(x, p) = \frac{p^2}{2m} + \frac{1}{2} m \omega^2 x^2$$

where m and ω are respectively the mass and circular frequency of the oscillator

(a) Determine the coefficient C .

(b) Find the averages $\langle p^2 \rangle$, $\langle x^2 \rangle$ and then $\langle E \rangle$, the average energy of an oscillator.

(c) Find $\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2}$, the uncertainty (or fluctuation) of the position x .

Solution: (a) From

$$\begin{aligned} 1 &= \int \frac{dn}{N} = C \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \exp\left(-\frac{E(x, p)}{k_B T}\right) dx dp = C \int_{-\infty}^{+\infty} e^{-\frac{m\omega^2 x^2}{2k_B T}} dx \int_{-\infty}^{+\infty} e^{-\frac{p^2}{2mk_B T}} dp \\ &= C \sqrt{\frac{2k_B T}{m\omega^2}} \int_{-\infty}^{+\infty} e^{-u^2} du \sqrt{2mk_B T} \int_{-\infty}^{+\infty} e^{-v^2} dv = C \frac{2\pi k_B T}{\omega}, \end{aligned}$$

we have

$$C = \frac{\omega}{2\pi k_B T}.$$

(b)

$$\begin{aligned}
\langle p^2 \rangle &= \frac{\int p^2 \frac{dn}{N}}{\int \frac{dn}{N}} = \frac{\int_{-\infty}^{+\infty} e^{-\frac{m\omega^2 x^2}{2k_B T}} dx \int_{-\infty}^{+\infty} p^2 e^{-\frac{p^2}{2mk_B T}} dp}{\int_{-\infty}^{+\infty} e^{-\frac{m\omega^2 x^2}{2k_B T}} dx \int_{-\infty}^{+\infty} e^{-\frac{p^2}{2mk_B T}} dp} \\
&= \frac{\int_{-\infty}^{+\infty} p^2 e^{-\frac{p^2}{2mk_B T}} dp}{\int_{-\infty}^{+\infty} e^{-\frac{p^2}{2mk_B T}} dp} = 2mk_B T \frac{\int_{-\infty}^{+\infty} u^2 e^{-u^2} du}{\int_{-\infty}^{+\infty} e^{-u^2} du} \\
&= 2mk_B T \frac{\int_0^{+\infty} e^{-t} t^{\frac{1}{2}} dt}{\int_0^{+\infty} e^{-t} t^{-\frac{1}{2}} dt} = 2mk_B T \frac{\Gamma\left(\frac{3}{2}\right)}{\Gamma\left(\frac{1}{2}\right)} = mk_B T. \\
\langle x^2 \rangle &= \frac{\int x^2 \frac{dn}{N}}{\int \frac{dn}{N}} = \frac{\int_{-\infty}^{+\infty} x^2 e^{-\frac{m\omega^2 x^2}{2k_B T}} dx \int_{-\infty}^{+\infty} e^{-\frac{p^2}{2mk_B T}} dp}{\int_{-\infty}^{+\infty} e^{-\frac{m\omega^2 x^2}{2k_B T}} dx \int_{-\infty}^{+\infty} e^{-\frac{p^2}{2mk_B T}} dp} \\
&= \frac{\int_{-\infty}^{+\infty} x^2 e^{-\frac{m\omega^2 x^2}{2k_B T}} dx}{\int_{-\infty}^{+\infty} e^{-\frac{m\omega^2 x^2}{2k_B T}} dx} = \frac{2k_B T}{m\omega^2} \frac{\int_{-\infty}^{+\infty} v^2 e^{-v^2} dv}{\int_{-\infty}^{+\infty} e^{-v^2} dv} \\
&= \frac{2k_B T}{m\omega^2} \frac{\int_0^{+\infty} e^{-t} t^{\frac{1}{2}} dt}{\int_0^{+\infty} e^{-t} t^{-\frac{1}{2}} dt} = \frac{2k_B T}{m\omega^2} \frac{\Gamma\left(\frac{3}{2}\right)}{\Gamma\left(\frac{1}{2}\right)} = \frac{k_B T}{m\omega^2}. \\
\langle E \rangle &= \left\langle \frac{p^2}{2m} + \frac{1}{2} m\omega^2 x^2 \right\rangle = \frac{\langle p^2 \rangle}{2m} + \frac{1}{2} m\omega^2 \langle x^2 \rangle \\
&= \frac{1}{2} k_B T + \frac{1}{2} k_B T = k_B T.
\end{aligned}$$

(c)

$$\begin{aligned}
\langle x \rangle &= \frac{\int x \frac{dn}{N}}{\int \frac{dn}{N}} = \frac{\int_{-\infty}^{+\infty} x e^{-\frac{m\omega^2 x^2}{2k_B T}} dx \int_{-\infty}^{+\infty} e^{-\frac{p^2}{2mk_B T}} dp}{\int_{-\infty}^{+\infty} e^{-\frac{m\omega^2 x^2}{2k_B T}} dx \int_{-\infty}^{+\infty} e^{-\frac{p^2}{2mk_B T}} dp} \\
&= \frac{\int_{-\infty}^{+\infty} x e^{-\frac{m\omega^2 x^2}{2k_B T}} dx}{\int_{-\infty}^{+\infty} e^{-\frac{m\omega^2 x^2}{2k_B T}} dx} = 0. \\
\Delta x &= \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \sqrt{\frac{k_B T}{m\omega^2}}.
\end{aligned}$$